Group II Row IV: the compact form, the running coupling, and why line 2 is not a running-coupling term

Conventions: $a \equiv \frac{k^+}{\vee}$ (kept exact), $\alpha_s = \frac{g^2}{4\pi}$, so $\frac{g^2}{8\pi^2} = \frac{\alpha_s}{2\pi}$ and $\frac{g^2 N_c}{4\pi^2} = \frac{\alpha_s}{2\pi} 2N_c$. $\beta_0 = \frac{11}{3} N_c$ (no quark loops in this row). $c_0 = 2e^{-\gamma_E}$. $d_\perp = 2 - \epsilon$. $\mathbf{k} \neq 0$ (measured gluon). Blue = added/changed, red = removed, relative to your compact form.

1 Definitions

Your $\frac{d^3 N_{\xi,\mathrm{LO}}}{d^3 k}$ with a weight f placed *inside* the integral:

$$\mathrm{LO}_\xi[f] \equiv \frac{1}{(2\pi)^3} \frac{g^2}{2\pi^2} \frac{1}{k^+} \int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}} f \, \frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2} \Big[U^{e'c}(y') - U^{e'c}(x')\Big] \, [U^{ec}(y) - U^{ce}(x)] \, \rho^{e'}(x') \rho^e(x), \tag{1.1}$$

$$\mathrm{LO}^{(y)}_\xi[f] \equiv \frac{1}{(2\pi)^3} \frac{g^2}{2\pi^2} \frac{1}{k^+} \int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}} f \, \frac{(x-y)\cdot(x'-y)}{(x-y)^2(x'-y)^2} \Big[U^{e'c}(y) - U^{e'c}(x')\Big] \, [U^{ec}(y) - U^{ce}(x)] \, \rho^{e'}(x') \rho^e(x), \tag{1.2}$$

$$\mathrm{LO}^{\mathrm{lp}}_\xi \equiv \frac{1}{(2\pi)^3} \frac{g^2}{2\pi^2} \frac{1}{k^+} \int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}} \left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2} - d_\perp \frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2} \right]$$
$$\times \Big[U^{e'c}(y') - U^{e'c}(x')\Big] \, [U^{ec}(y) - U^{ce}(x)] \, \rho^{e'}(x') \rho^e(x). \tag{1.3}$$

$\mathrm{LO}_\xi \equiv \mathrm{LO}_\xi[1] = \frac{1}{\xi} \frac{d^3 N_{\mathrm{LO}}}{d^3 k}\big(\frac{k^+}{\xi}, \frac{\mathbf{k}}{\xi}\big)$ and $\mathrm{LO} \equiv \mathrm{LO}_{\xi=1} = \frac{d^3 N_{\mathrm{LO}}}{d^3 k}$. $\mathrm{LO}^{(y)}_\xi$ is LO_ξ at $y' = y$ in the kernel and in $U(y')$ (your line 2). H_4 is your line 3 without $-(1 - \frac{k^+}{\vee})$:

$$H_4 \equiv \frac{1}{(2\pi)^3} \frac{g^4 N_c}{8\pi^5} \frac{1}{k^+} \int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)} \frac{(w-z)\cdot(w'-z)}{(w-z)^2(w'-z)^2} \frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2} \Big[U^{e'c}(z) - U^{e'c}(x')\Big] \, [U^{ec}(z) - U^{ce}(x)] \, \rho^{e'}(x') \rho^e(x). \tag{1.4}$$

$$\ell_\Delta(\mu) \equiv 2\gamma_E + \log \frac{(y'-y)^2 \mu^2}{4} = \log \frac{(y'-y)^2 \mu^2}{c_0^2}, \tag{1.5}$$

$$P_{gg}(\xi) = 2N_c \Big[\frac{\xi}{(1-\xi)_+} + \frac{1-\xi}{\xi} + \xi(1-\xi) \Big] + \frac{\beta_0}{2} \delta(1-\xi), \qquad P^{\mathrm{real}}_{gg}(\xi) \equiv P_{gg}(\xi) - \frac{\beta_0}{2} \delta(1-\xi), \tag{1.6}$$

with the plus distribution of $[0,1]$: $\int_a^1 d\xi \frac{g(\xi)}{(1-\xi)_+} = \int_a^1 d\xi \frac{g(\xi)-g(1)}{1-\xi} + g(1)\log(1-a)$.

2 Is the compact form correct?

Line	Status	Comment
1	**fix**	(i) The bracket depends on $y' - y$: it must be the weight f *inside* $\mathrm{LO}_\xi[f]$, not a factor outside $\frac{d^3 N_{\xi,\mathrm{LO}}}{d^3 k}$. (ii) The colour factor written after line 1 is already inside $\frac{d^3 N_{\xi,\mathrm{LO}}}{d^3 k}$ (your own definition): remove it. (iii) $\mathcal{P}_{gg}$ must be the full P_{gg} of (1.6): standard plus distribution on $[0,1]$ (cut at $\xi \geq a$) and $+\frac{\beta_0}{2}\delta(1-\xi)$. Then $\mathcal{P}_{gg} - \frac{\beta_0}{2}\delta = P^{\mathrm{real}}_{gg}$ reproduces your original ξ-bracket *and* your $\log(1 - \frac{k^+}{\vee})$ line. With your prescription $[\frac{1}{p^+}]_+$ instead, add $-\log(1-a)\,\delta(1-\xi)$. (iv) $-2\gamma_E - \log(\dots)$ is the $\epsilon \to 0$ form of $-\frac{d_\perp}{2}\ell_\Delta$; see (2.1).
2	correct	Same as the original line. Its $-\log$ is $-\frac{d_\perp}{2}\log$ at $\epsilon \to 0$ (no pole multiplies it). $2\gamma_E$ may be added or not: it multiplies $\delta^2(\mathbf{k})$, see Sec. 5.
3	correct	$-(1 - \frac{k^+}{\vee}) H_4$.
4	correct	$\frac{\alpha_s}{2\pi} 2N_c \int_a^1 d\xi \frac{1-\xi}{\xi} \mathrm{LO}^{\mathrm{lp}}_\xi$; $d_\perp \to 2$ at the end (finite).

The exact bracket, with $d_\perp = 2 - \epsilon$ (checked in `checks.py`):

$$-\frac{d_\perp}{\epsilon} - d_\perp \gamma_E - \frac{d_\perp}{2} \log \frac{(y'-y)^2\mu^2}{4} - 1 = -\frac{2}{\epsilon} - \frac{d_\perp}{2}\ell_\Delta(\mu) = -\frac{2}{\epsilon} - \ell_\Delta(\mu) + \frac{\epsilon}{2}\ell_\Delta(\mu). \tag{2.1}$$

The last term has no pole in front of it and vanishes at $\epsilon \to 0$.

Corrected compact form.

$$\frac{d^3N}{d^3k} = \frac{\alpha_s}{2\pi}\int_a^1 \frac{d\xi}{\xi^2}\Big[P_{gg}(\xi) - \frac{\beta_0}{2}\delta(1-\xi)\Big]\,\mathrm{LO}_\xi\Big[-\frac{2}{\epsilon} - \frac{d_\perp}{2}\ell_\Delta(\mu_{\overline{\rm MS}})\Big] \times\big[U^{e'c}(y') - U^{e'c}(x')\big]\big[U^{ec}(y) - U^{ce}(x)\big]\rho^{e'}(x')\rho^{e}(x)$$
$$+ \frac{\alpha_s}{2\pi}\,2N_c\int_a^1 \frac{d\xi}{\xi^2}\frac{1-\xi}{\xi}\,\mathrm{LO}_\xi^{(y)}\Big[\frac{d_\perp}{2}\log\frac{(y'-y)^2\mu^2_{\overline{\rm MS}}}{4}\Big] - \Big(1-\frac{k^+}{\vee}\Big)H_4 + \frac{\alpha_s}{2\pi}2N_c\int_a^1 d\xi\frac{1-\xi}{\xi}\mathrm{LO}_\xi^{\rm lp}. \tag{2.2}$$

3 Simplified: DGLAP + running coupling + finite

Step 1: lines 1 and 2 as one ξ integral. Line 2 has the same $\frac{1-\xi}{\xi}$ as the small-ξ part of P_{gg}, so

$$(1)+(2) = \frac{\alpha_s}{2\pi}\int_a^1\frac{d\xi}{\xi^2}\Big\{P^{\rm real}_{gg}(\xi)\,\mathrm{LO}_\xi\Big[-\frac{2}{\epsilon}-\frac{d_\perp}{2}\ell_\Delta(\mu)\Big] + 2N_c\frac{1-\xi}{\xi}\,\mathrm{LO}_\xi^{(y)}\Big[\frac{d_\perp}{2}\ell_\Delta(\mu)\Big]\Big\}. \tag{3.1}$$

For $\xi\to 0$ the phase $e^{-ik\cdot(y'-y)/\xi}$ forces $y'\to y$, so the log-part of $\frac{1-\xi}{\xi}\big(\mathrm{LO}_\xi - \mathrm{LO}_\xi^{(y)}\big)$ vanishes there: no log $\frac{\vee}{k^+}$ is produced by (3.1) (it is already in BFKL).

Step 2: split $P^{\rm real}_{gg} = P_{gg} - \frac{\beta_0}{2}\delta(1-\xi)$. At $\xi=1$, $\mathrm{LO}_\xi[f]=\mathrm{LO}[f]$:

$$(3.1) = \underbrace{\frac{\alpha_s}{2\pi}\int_a^1\frac{d\xi}{\xi^2}\Big\{P_{gg}\,\mathrm{LO}_\xi\Big[-\frac{2}{\epsilon}-\frac{d_\perp}{2}\ell_\Delta(\mu)\Big] + 2N_c\frac{1-\xi}{\xi}\mathrm{LO}_\xi^{(y)}\Big[\frac{d_\perp}{2}\ell_\Delta(\mu)\Big]\Big\}}_{\text{DGLAP}} + \underbrace{\frac{\alpha_s\beta_0}{4\pi}\,\mathrm{LO}\Big[\frac{2}{\epsilon}+\frac{d_\perp}{2}\ell_\Delta(\mu)\Big]}_{\text{running coupling}}. \tag{3.2}$$

Step 3: DGLAP. The pole $\frac{\alpha_s}{2\pi}(-\frac{2}{\epsilon})P_{gg}\otimes\mathrm{LO}$ is removed by the $\overline{\rm MS}$ fragmentation function, $D^{\rm bare} = D(\mu_F) + \frac{\alpha_s}{2\pi}\frac{2}{\epsilon}P_{gg}\otimes D$, with $\frac{d^3N_h}{d^3k} = \int\frac{d\zeta}{\zeta^3}D(\zeta)\frac{d^3N_g}{d^3k}(\frac{k}{\zeta})$. Setting $\mu=\mu_F$:

$$\boxed{\left.\frac{d^3N}{d^3k}\right|_{\rm DGLAP} = -\frac{\alpha_s}{2\pi}\int_a^1\frac{d\xi}{\xi^2}\Big\{P_{gg}(\xi)\,\mathrm{LO}_\xi\Big[\ell_\Delta(\mu_F)\Big] - 2N_c\frac{1-\xi}{\xi}\,\mathrm{LO}_\xi^{(y)}\Big[\ell_\Delta(\mu_F)\Big]\Big\}} \tag{3.3}$$

($\frac{d_\perp}{2}\to 1$: no pole is left in front). Check: $\mu_F^2\partial_{\mu_F^2}$ of (3.3) is $-\frac{\alpha_s}{2\pi}P_{gg}\otimes\mathrm{LO}$, which cancels $\mu_F^2\partial_{\mu_F^2}D = \frac{\alpha_s}{2\pi}P_{gg}\otimes D$. The $\mathrm{LO}^{(y)}$ term contributes nothing to this derivative (Sec. 5).

Step 4: running coupling (details in Sec. 4):

$$\boxed{\left.\frac{d^3N}{d^3k}\right|_{\rm RC} = \frac{\alpha_s\beta_0}{4\pi}\,\mathrm{LO}\Big[\log\frac{(y'-y)^2\mu_R^2}{c_0^2}\Big]} \tag{3.4}$$

Step 5: finite.

$$\boxed{\left.\frac{d^3N}{d^3k}\right|_{\rm finite} = -\Big(1-\frac{k^+}{\vee}\Big)H_4 + \frac{\alpha_s}{2\pi}\,2N_c\int_a^1 d\xi\,\frac{1-\xi}{\xi}\,\mathrm{LO}_\xi^{\rm lp}\Big|_{d_\perp=2}} \tag{3.5}$$

So $\dfrac{d^3N_{\rm G2\,IV}}{d^3k} = (3.3)+(3.4)+(3.5)$.

4 Running coupling, step by step

1. **Source.** Only the $-\frac{\beta_0}{2}\delta(1-\xi)$ in line 1: $\frac{\alpha_s}{2\pi}(-\frac{\beta_0}{2})\mathrm{LO}\big[-\frac{2}{\epsilon}-\frac{d_\perp}{2}\ell_\Delta(\mu)\big] = \frac{\alpha_s\beta_0}{4\pi}\mathrm{LO}\big[\frac{2}{\epsilon}+\frac{d_\perp}{2}\ell_\Delta(\mu)\big]$.
2. **Coupling counterterm.** LO $\propto g^2$. In $\overline{\rm MS}$, $\alpha_s^{\rm bare}\mu^\epsilon = \alpha_s(\mu_R)\big[1-\frac{\alpha_s\beta_0}{4\pi}\frac{2}{\epsilon}\big]$ (here $\frac{1}{\epsilon_{d=4-2\epsilon'}} = \frac{2}{\epsilon}$), so the LO term gives $-\frac{\alpha_s\beta_0}{4\pi}\frac{2}{\epsilon}\mathrm{LO}$.
3. **Sum** at $\mu=\mu_R$: the pole cancels, and $\frac{d_\perp}{2}\to 1$: $\frac{\alpha_s\beta_0}{4\pi}\mathrm{LO}\big[\ell_\Delta(\mu_R)\big] = \frac{\alpha_s\beta_0}{4\pi}\mathrm{LO}\big[\log\frac{(y'-y)^2\mu_R^2}{c_0^2}\big]$.
4. **Resummed.** With $\alpha_s(Q) = \alpha_s(\mu_R)\big[1-\frac{\alpha_s\beta_0}{4\pi}\log\frac{Q^2}{\mu_R^2}\big]$,

$$\mathrm{LO}\Big|_{\alpha_s(\mu_R)} + \frac{\alpha_s\beta_0}{4\pi}\mathrm{LO}\Big[\log\frac{(y'-y)^2\mu_R^2}{c_0^2}\Big] = \mathrm{LO}\Big|_{\alpha_s\to\alpha_s\left(\frac{c_0}{|y'-y|}\right)} + O(\alpha_s^3), \tag{4.1}$$

   i.e. the coupling of the emitted gluon runs with the amplitude–conjugate separation $|y'-y|$, inside the y,y' integrals.
5. **Check.** $\mu_R^2\partial_{\mu_R^2}$: LO gives $-\frac{\alpha_s^2\beta_0}{4\pi}$, (3.4) gives $+\frac{\alpha_s^2\beta_0}{4\pi}$: μ_R-independent.
6. With quarks, $\beta_0\to\frac{11}{3}N_c - \frac{2}{3}n_f$; the n_f part comes from quark-loop rows, not from this row.

(This uses $\mu=\mu_{\overline{\rm MS}}$ as in your equation, i.e. γ_E and $\log 4\pi$ already in μ; then the scale is $c_0/|y'-y|$, $c_0 = 2e^{-\gamma_E}$.)

5 Line 2 does *not* contribute to the running coupling

(a) Its μ-dependence is zero for $\mathbf{k} \neq 0$. In $\mathrm{LO}_\xi^{(y)}$ the kernel $\frac{(x-y)\cdot(x'-y)}{(x-y)^2(x'-y)^2}$ and $U(y)$ do not depend on y'. The only y' dependence is the phase and the weight. Write the weight as $\log(y'-y)^2 + \log\frac{\mu^2}{4}$ (and $2\gamma_E$, $-\frac{2}{\epsilon}$ if added). The constant part gives

$$\Big[\log\frac{\mu^2}{4} + \text{const}\Big]\int d^2y'\, e^{-ik\cdot(y'-y)/\xi} = \Big[\log\frac{\mu^2}{4} + \text{const}\Big](2\pi)^2\xi^2\,\delta^2(\mathbf{k}) = 0 \qquad (\mathbf{k}\neq 0). \tag{5.1}$$

So line 2 has **no μ**, **no pole**, and hence nothing to renormalize the coupling. Only $\log(y'-y)^2$ survives; from $\nabla^2 \log r^2 = 4\pi\delta^2(r)$,

$$\int d^2y'\, e^{-ik\cdot(y'-y)/\xi}\log(y'-y)^2 = -\frac{4\pi\xi^2}{\mathbf{k}^2} \qquad (\mathbf{k}\neq 0), \tag{5.2}$$

which is finite and μ-independent.

(b) Its value. With $\int_a^1 d\xi\,\frac{1-\xi}{\xi^3}\,\xi^2 = \log\frac{\vee}{k^+} - \big(1-\frac{k^+}{\vee}\big)$ and $\big|\int_w e^{ik\cdot w}\frac{(w-z)^i}{(w-z)^2}\big|^2 = \frac{4\pi^2}{\mathbf{k}^2}$ (normalisations checked in `checks.py`),

$$\text{line 2} = -\Big[\log\frac{\vee}{k^+} - \Big(1-\frac{k^+}{\vee}\Big)\Big]H_4. \tag{5.3}$$

This is the $y'=y$ subtraction of the small-ξ ($\frac{1-\xi}{\xi}$) part of line 1: line 1 contains $+\log\frac{\vee}{k^+}H_4$ from $\xi\to 0$, and (5.3) removes it. That is a rapidity (BFKL) log, not a UV/coupling log. That is why it is kept inside the DGLAP integral in (3.3), where the two cancel and no $\log\frac{\vee}{k^+}$ is shown.

(c) Wrong structure for running coupling. A running-coupling term must be $\propto \beta_0\,\delta(1-\xi)\times\mathrm{LO}\times\log\mu_R^2$: it multiplies LO itself, at $\xi=1$. Line 2 has the colour weight $2N_c\frac{1-\xi}{\xi}$, integrated over ξ, and multiplies $\mathrm{LO}_\xi^{(y)}$, not LO.

(d) Consistency. Line 1 alone already gives the full μ-dependence: $-\frac{\alpha_s}{2\pi}P_{gg}\otimes\mathrm{LO}$ (cancels DGLAP of D) and $+\frac{\alpha_s\beta_0}{4\pi}\mathrm{LO}$ (cancels the running of α_s in LO). A μ-term from line 2 would spoil both.

Caveat. Only if $\mathbf{k}$ is integrated over (so $\delta^2(\mathbf{k})$ does not vanish) would line 2's constant part matter. For the measured $\mathbf{k}\neq 0$ it does not.